Topic:-

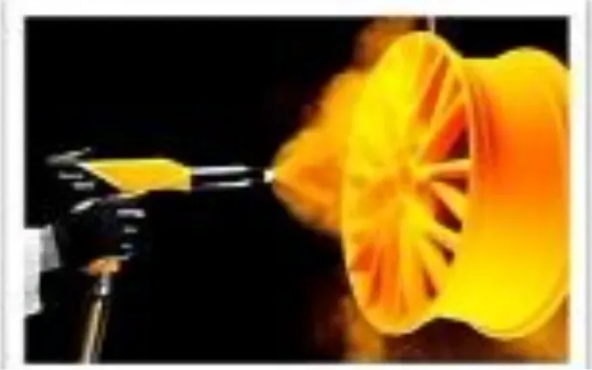
Visit the places where powder coating, Teflon coating is done.
Get information about the process and present it in the class.

Introduction:-

Powder coating and Teflon coating are modern techniques used to protect and improve the surface of different materials, especially metals. These coatings help in increasing durability, preventing rust, and giving a smooth and attractive finish. Powder coating uses dry powder applied electrostatically and then heated to form a hard layer, while Teflon coating provides a non-stick, heat-resistant surface commonly used in cookware and industrial equipment.

Visiting places where these coatings are done helps us understand the real-life application of these processes, the machines involved, and their importance in industries. This project aims to study the working, advantages, and uses of powder coating and Teflon coating in a practical way.

Powder Coating:-



Powder coating is a dry finishing process that involves the application of a free-flowing, dry powder to a surface. The powder is typically made of a combination of resin, pigment, and other additives, and is applied using an electrostatic gun. The gun charges the powder particles with an electric charge, which causes them to stick to the surface being coated.The coated surface Is then heated in an oven to melt and cure the powder, creating a hard, durable finish. The powder coating process offers several advantages over other coating methods, including excellent durability, scratch resistance, and the ability to produce a wide range of colors and finishes.

Teflon Coating:-

Teflon coating, also known as PTFE coating, is a type of fluoropolymer coating that is used to provide a non-stick surface on a variety of materials. Teflon is a brand name for PTFE, which stands for polytetrafluoroethylene. This material has excellent non-stick properties and is resistant to chemicals, heat, and abrasion.
The Teflon coating process typically involves spraying a liquid solution of PTFE onto the surface being coated, then curing the coating at high

temperatures to create a solid, non-stick surface. Teflon coatings are commonly used on cookware, industrial equipment, and various other applications where a non-stick surface is required.

Advantages:-

1. Provides strong protection against rust and corrosion
2. Increases durability of metal surfaces
3. Gives smooth and attractive finishing
4. Powder coating is eco-friendly (less pollution)

Disadvantages :-

1. Costly equipment
2. Requires skilled workers
3. Teflon may release harmful fumes at very high temperature
4. Limited repair options

Conclusion:-

Powder coating and Teflon coating are important surface finishing techniques used in many industries. They improve the strength, appearance, and life of materials, especially metals. Powder coating provides a durable and attractive finish, while Teflon coating offers non-stick and heat-resistant properties. Visiting coating centers helped in understanding the real-life process, machines used, and safety measures followed. These methods are efficient, long-lasting, and environmentally friendly compared to traditional painting. Overall, this project helped in gaining practical knowledge about modern coating technologies and their wide applications in daily life and industries.

Slogan:-

1. "Strong coating, long-lasting protection!"
2. "Coat it right, make it bright!"

3. “Better coating, better life!”
4. “Protect today, last tomorrow!”
5. “Smooth finish, stronger future!”

References:-

1.NCERT Chemistry Textbook (Class 12)
2.Information collected during visit to local coating center
3.https://www.sciencedirect.com
4.https://www.tiger-coatings.com
5.https://www.coatingsolutions.com
6.YouTube educational videos on powder coating and Teflon coating